Q1: Data Cleaning using Duplicate Detection (Patient Registration)

Step 1: Original Records with Duplicate Flag

S.No	Patient ID	Occurrence Count (in list)	Duplicate? (Y/N)
1	P101	1	No
2	P102	1	No
3	P105	2	Yes
4	P103	2	Yes
5	P105	2	Yes
6	P107	1	No
7	P108	1	No
8	P103	2	Yes
9	P110	1	No
10	P112	1	No

Step 2: Unique Patient List (duplicates removed)

S.No	Unique Patient ID
1	P101
2	P102
3	P105
4	P103
5	P107
6	P108
7	P110
8	P112

Step 3: Summary

Total Records Extracted:		**10**
Duplicate ID Values Found:	**P105, P103**	
Number of Duplicate (extra) entri		**2**
Number of Unique Patients:		**8**

Q2: Covariance Analysis between Treatment Cost and Hospital Stay

S.No	reatment Cost x (₹ '000	ospital Stay y (day	x - mean(x)	y - mean(y)	(x-mean x) * (y-mean y)
1	20	3	-10	-2.2	22
2	25	4	-5	-1.2	6
3	30	5	0	-0.2	0
4	35	6	5	0.8	4
5	40	8	10	2.8	28
	Sum ->				**60**
Mean (x, y) ->	**30**	**5.2**			

Covariance Calculation

Population Covariance $Cov(x,y) = \Sigma[(x-\bar{x})(y-\bar{y})] / n$	**12**
Sample Covariance $Cov(x,y) = \Sigma[(x-\bar{x})(y-\bar{y})] / (n-1)$	**15**

Interpretation

The covariance between treatment cost and hospital stay is positive (Cov ≈ +15, sample basis / +12, population basis). This indicates a positive relationship: as the treatment cost increases, the duration of hospital stay also tends to increase. Patients with higher treatment costs are generally associated with longer hospital stays, which is clinically intuitive since more severe/complex cases require both longer stays and more expensive treatment.

Q3: Outlier Detection using Z-Score (Patient Recovery Time)

S.No	Recovery Time (days	x - mean	(x-mean)^2	Z-score = (x-mean)/std
1	8	-5.545454545	30.75206612	-0.7837277069
2	9	-4.545454545	20.66115702	-0.6423997597
3	10	-3.545454545	12.57024793	-0.5010718126
4	10	-3.545454545	12.57024793	-0.5010718126
5	11	-2.545454545	6.479338843	-0.3597438655
6	12	-1.545454545	2.388429752	-0.2184159183
7	12	-1.545454545	2.388429752	-0.2184159183
8	13	-0.5454545455	0.2975206612	-0.07708797117
9	14	0.4545454545	0.2066115702	0.06423997597
10	15	1.454545455	2.115702479	0.2055679231
11	35	21.45454545	460.2975207	3.032126866
	Sum of squares ->		**550.7272727**	
Mean:	**13.54545455**			
Standard Deviation (popul	**7.075741354**			

Outlier Identification (|z| > 3)

Observation	Z-score	Outlier (|z|>3)
35	3.032126866	Yes - Outlier

Conclusion: Only the value 35 has |z| > 3, so it is identified as the sole outlier in the dataset.

Q4: Data Discretization using Equal-Width Binning (Blood Glucose l

Bin Width Calculation

Minimum v	**75**
Maximum	**145**
Range (Ma	**70**
Number of	**3**
Bin width (	**23.33333333**

Bin Boundaries

Bin	Lower Boundary	pper Boundar
Bin 1	75	98.33333333
Bin 2	98.33333333	121.6666667
Bin 3	121.6666667	145

Transformed Dataset (Smoothing by Bin Boundaries — each value replaced by neares

S.No	Original Value	Assigned Bin	learest Lower Edge Dis	learest Upper Edge Dis	Transformed Value
1	75	Bin 1	0	23.33333333	75
2	82	Bin 1	7	16.33333333	75
3	88	Bin 1	13	10.33333333	98.33333333
4	92	Bin 1	17	6.333333333	98.33333333
5	95	Bin 1	20	3.333333333	98.33333333
6	99	Bin 2	0.6666666667	22.66666667	98.33333333
7	104	Bin 2	5.666666667	17.66666667	98.33333333
8	110	Bin 2	11.66666667	11.66666667	98.33333333
9	118	Bin 2	19.66666667	3.666666667	121.6666667
10	122	Bin 3	0.3333333333	23	121.6666667
11	130	Bin 3	8.333333333	15	121.6666667
12	145	Bin 3	23.33333333	0	145

Q5: Standardization of Insurance Claim Amounts

S.No	Claim Amount (₹ '000	x - mean	(x-mean)^2	-score = (x-mean)/std
1	12	-12.5	156.25	-1.485046613
2	15	-9.5	90.25	-1.128635426
3	18	-6.5	42.25	-0.772224239
4	20	-4.5	20.25	-0.5346167808
5	22	-2.5	6.25	-0.2970093227
6	25	0.5	0.25	0.05940186453
7	28	3.5	12.25	0.4158130517
8	30	5.5	30.25	0.6534205099
9	35	10.5	110.25	1.247439155
10	40	15.5	240.25	1.841457801
	Sum of squares ->		**708.5**	
Mean:	**24.5**			
Standard	**8.417244205**			

Claims Above One Standard Deviation from the Mean (z > 1, i.e. value > mean + 1×

Upper Thr	**32.9172442**

S.No	Claim Amount	Z-score	bove 1 SD? (z>1
1	12	-1.485046613	No
2	15	-1.128635426	No
3	18	-0.772224239	No
4	20	-0.5346167808	No
5	22	-0.2970093227	No
6	25	0.0594018645	No
7	28	0.4158130517	No
8	30	0.6534205099	No
9	35	1.247439155	Yes
10	40	1.841457801	Yes